# What Should a Joint-Embedding Predictor Predict? Target Composition, Context Loss, and Equity in Medical Image SSL

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## Abstract

Masking policy is an under-audited design choice in medical self-supervision. We study patch-level I-JEPA pretraining for glaucoma classification from optical coherence tomography (OCT). We found no prior report of the following medical-imaging consequence of stock I-JEPA batching: variable-length context masks are truncated to the batch minimum by retaining a row-major prefix, systematically deleting lower image rows. In our rectangle-based arms this removes 31–36% of sampled context and leaves up to 11.02% of slices with no retinal anatomy delivered to the encoder. Across four epoch-50 arms, frozen probe AUC is non-monotonic in target anatomy purity (31.6%→0.8641, 39.7%→0.8740, 43.2%→0.8761, 97.5%→0.8654). The near-pure arm's predictor error rises 2.76× while its relative reliance on anatomy context falls, supporting predictor collapse rather than anatomy starvation. Post-hoc equity analysis finds Black patients worst served in all 18 valid probes; racial AUC gaps vary 1.97×, whereas the severe-to-mild disease gap remains 0.1286–0.1394. These are single-seed, observational results: 18 probes derive from six pretraining trajectories, the test split was consulted during earlier development so all findings are exploratory rather than confirmatory, and we do not claim that anatomy-shaped targets outperform rectangles.

## 1 Introduction

Masked and predictive self-supervision learns visual representations by withholding part of an image and predicting pixels, tokens, or latent representations [He et al., 2022, Bao et al., 2022, Assran et al., 2023]. The masking policy determines both what supervision is presented and what evidence remains visible. This choice is especially consequential in retinal OCT, where anatomy occupies a narrow band within many B-scans.

We audit I-JEPA [Assran et al., 2023] on the FairVision glaucoma cohort [Luo et al., 2024]. The programme began with the hypothesis that connected anatomy-shaped targets would beat anatomy-guided rectangles. The evidence rejects that framing: anatomy is ahead by 0.0054 at epoch 30, but the direction reverses by $-0.0107$ at epoch 50, and the early comparison uses different guide caches. We therefore study the more defensible question: *what effective prediction problem does a mask policy create after collation?*

Our contributions are:

- We trace a spatially biased context-erasure failure to stock-style batch-minimum prefix truncation and quantify its batch-size dependence.

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- We measure target and delivered-context composition, finding an observational, non-monotonic purity–AUC relationship rather than a universal purity optimum.
- We diagnose the near-pure arm's failure as predictor collapse, while refuting the simpler anatomy-starvation explanation.
- We show consistent racial ordering across saved probes and a policy-invariant mild-disease penalty, with statistical and independence limits stated explicitly.

# 2 Background and related work

**Joint-embedding prediction.** I-JEPA predicts target-encoder representations from a masked context view [Assran et al., 2023]. Subsequent work studies representation collapse and target ordering [Mo and Tong, 2024, He et al., 2025], but we found no prior report of the spatially biased context-erasure mode examined here. Our claim is search-qualified: we report a medical-imaging consequence of released I-JEPA-style mask batching for which we found no prior report, not a defect unique to our anatomy-guided code.

**Semantic and anatomical masking.** Attention-guided and reconstruction-guided methods establish that target content matters [Kakogeorgiou et al., 2022, Li et al., 2024]. Ceballos Arroyo et al. use a pretrained segmenter to guide masked autoencoding in medical images [Ceballos Arroyo et al., 2026]; related anatomical masking also predates this study [Huang et al., 2025]. We therefore do not claim the first anatomy-guided medical masking method. Our narrower contribution is an OCT/I-JEPA audit that measures target purity, delivered context, predictor health, and trajectory reversal.

**Ophthalmic equity.** Medical models can encode demographic signals and exhibit subgroup disparities [Gichoya et al., 2022, Jin et al., 2024]. FairVision was designed to support equity analysis in ophthalmic AI [Luo et al., 2024]. We use its released race and disease-severity attributes, but treat repeated probes from the same encoder as non-independent.

# 3 Experimental setup

**Data and permitted use.** We use the released FairVision glaucoma cohort. Its split contains 6,000 training, 1,000 validation, and 3,000 test OCT volumes. Each `oct_bscans` array is a $200 \times 200 \times 200$ uint8 B-scan stack. Evaluation samples 100 of the 200 B-scans per volume. The test set has 1,466 positive and 1,534 negative volumes. The data license is CC BY-NC-ND 4.0 and permits non-commercial research, not clinical use.

**Encoder and probing.** We use patch-level I-JEPA with a ViT-B/16 encoder [Dosovitskiy et al., 2021]. Slices are processed on a $16 \times 16$ patch grid. Unless stated otherwise, evaluation freezes the encoder, mean-pools patch features, and fits a linear glaucoma head with probe seed 42. Every masking arm descends from the same epoch-25 checkpoint, whose test AUC is 0.8487. This shared ancestor improves comparability but does not replace pretraining-seed replication.

**Mask policies and measurements.** Random uses standard multiblock rectangles; oracle places rectangles using a hand-crafted retinal band; envelope places rectangles using frozen MIRAGE segmentation [Morano et al., 2025]; COVER greedily places rectangles while enforcing an anatomy-visibility floor; and blob uses connected, near-pure anatomy targets. For each slice we measure anatomy recall by targets, target purity, anatomy retained in delivered context, target/context budgets, and the zero-anatomy context rate.

# 4 Spatially biased context erasure

## 4.1 Mechanism

The defect is a three-link causal chain in the mask collator:

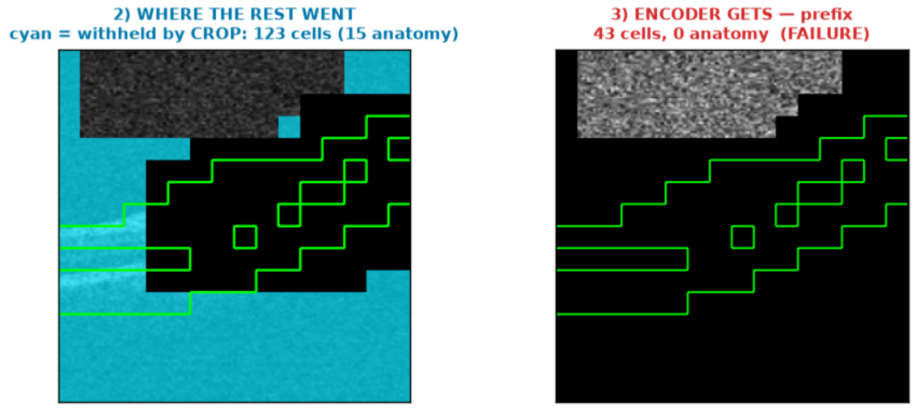


Figure 1: Measured retained-versus-discarded encoder context for a stored failing slice. Cyan marks 123 non-target cells not delivered to the encoder, including 15 anatomy cells; the encoder receives 43 smallest-index cells from the top rows and zero anatomy. The total withheld count includes pre-collation context selection as well as batch truncation. Green traces the retinal guide. This panel is cropped from the stored diagnostic image.

```
self.height = input_size[0] // patch_size
self.width = input_size[1] // patch_size
indices.append(r * self.width + c)
masks_enc[e].append(
    torch.tensor(sorted(best_indices), dtype=torch.long)
)
min_len = min(t.numel() for t in group)
torch.stack([t[:min_len] for t in group], dim=0)
```

First, $r \times \texttt{width} + c$ is row-major: every index in a lower image row exceeds every index in a higher row. Second, encoder masks are explicitly sorted ascending before storage. Third, `_truncate_and_stack` retains `t[:min_len]`. It therefore keeps the smallest indices—the top rows—and discards the largest indices—the bottom rows. The row-major ascending ordering, created by `_block_to_indices` and redundantly re-enforced by `sorted()` at encoder storage, is what makes this excision deterministic and spatially systematic rather than a random subset. In OCT B-scans the retinal band occupies a limited vertical extent, so this is a systematic anatomical excision, not benign padding (Figure 1).

This is a faithful port of released upstream behavior (I-JEPA commit `52c1ae95d05f743e000e8f10a1f3a79b10cff048`, mask-collator lines 145–175), not a local defect introduced by our anatomy code. We are not aware of a prior report of this failure mode. In our rectangle arms at batch size 64 it removes 31–36% of sampled context, including for plain random masks.

The proof above applies only to encoder/context masks. Predictor masks are stored without the `sorted()` call and follow a separate, more aggressive path: one global minimum is taken across every predictor group and sample before retaining `t[:global_min_pred]`. We therefore make no claim about the spatial character of predictor-mask truncation.

### 4.2 Failure accounting

The effect is anatomically consequential (Figures 1–2). In one failing slice, all 256 patches balance exactly as 90 targets, 123 withheld patches, and 43 patches delivered to the encoder. Its 65 anatomy patches balance as 50 targeted and 15 withheld, leaving zero anatomy visible. The withheld count includes pre-collation context sampling as well as batch truncation, so it is not attributed to truncation alone.

Repeated draws show that this is not an intrinsic "bad slice" property: 0% of slices are always blank, and 52.0% are never blank versus 44.6% expected under 12 independent draws at the aggregate

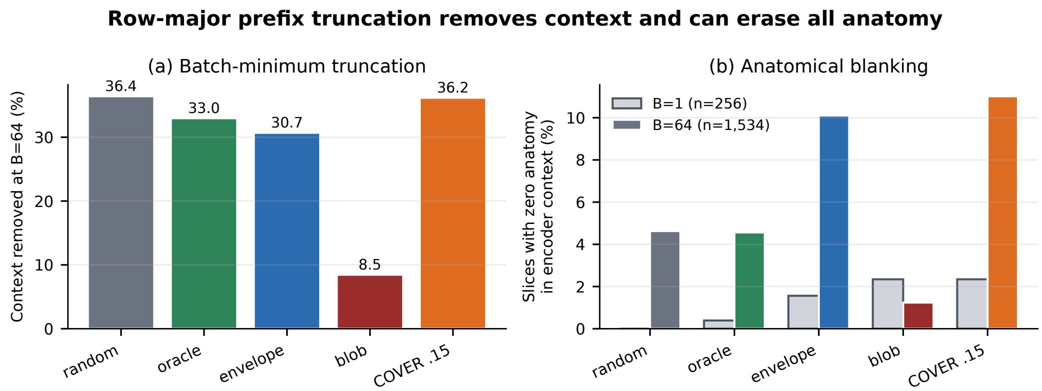


Figure 2: Batch-size audit of prefix truncation. Rectangle policies lose 31–36% of sampled context at $B = 64$. Zero-anatomy context increases from $B = 1$ to $B = 64$: random 0.00→4.63%, envelope 1.56→10.10%, and COVER $f = .15$ 2.34→11.02%. The audits use $n = 256$ and $n = 1{,}534$ slices, respectively, and are separate samples rather than a paired intervention. The independent $n = 6{,}137$ pass of Table 1 puts the same $B = 64$ rates at 3.68% and 8.07%; we report both passes rather than pooling them.

Table 1: Post-collation composition. Values are percentages except context cell counts and AUC. AUC uses the standard 100-slice, seed-42 frozen mean-pool linear protocol. Random, oracle, COVER, and envelope share an $n = 6{,}137$ audit; blob is a separate $n = 1{,}534$ pass.

| Arm | $n$ | Recall | Purity | Ctx. anat. | Anat. cells | Ctx. cells | Zero anat. | ep50 AUC |
|---|---|---|---|---|---|---|---|---|
| random | 6,137 | 53.04 | 31.58 | 26.38 | 18.25 | 69.09 | 3.68 | .8641 |
| oracle | 6,137 | 61.58 | 39.69 | 19.34 | 14.93 | 77.23 | 4.19 | .8740 |
| COVER $f = .21$ | 6,137 | 73.09 | 40.88 | 14.56 | 9.28 | 63.62 | 7.84 | – |
| envelope | 6,137 | 77.58 | 43.19 | 11.38 | 8.63 | 76.41 | 8.07 | .8761 |
| blob | 1,534 | 82.07 | 97.50 | 6.26 | 9.97 | 160.00 | 1.24 | .8654 |

6.5104% blank rate. The failure varies across draws; the 12-draw audit found no fixed always-blank subset, although blanking risk remains concentrated in some slices.

## 5 Target composition and downstream quality

**The hypothesis we started from.** Anatomy is where the diagnostic information is, and the predictor agrees with that intuition. Per removed context token, anatomy is worth 3.9–6.6× more to the predictor than background across every healthy checkpoint we measured (Figure 4a), and region-restricted probes order the same way, with anatomy-only pooling above background-only pooling in all four arms (Figure 4b). The natural design response is to spend the masking budget on anatomy, and our policies do so progressively: random rectangles place 31.6% of target cells on anatomy, oracle 39.7%, COVER 40.9%, envelope 43.2%, and blob 97.5%.

**The hypothesis fails at the extreme.** Table 1 separates what the sampler targets from what the encoder receives. At epoch 50, AUC increases from 0.8641 at 31.6% purity to 0.8740 and 0.8761 at 39.7% and 43.2%, then falls to 0.8654 at 97.5% (Figure 3). The same reversal appears against anatomy recall: performance improves as targets cover 53.0%, 61.6%, and 77.6% of slice anatomy, then drops at 82.1%. Among these four confounded, single-seed arms, the two highest observed AUCs occur at purities of 39.7% and 43.2%. Because four points on one dataset move several design axes together, this sparse comparison supports only rejection of a monotonic “more anatomy is always better” rule; it does not estimate a preferred range, a dose response, or a causal purity effect.

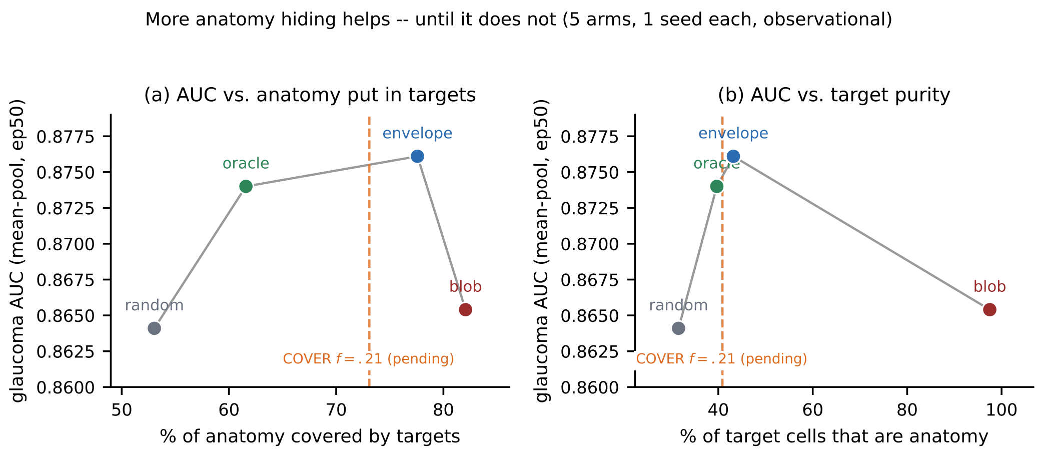


Figure 3: Downstream AUC is non-monotonic in how much anatomy the policy puts into targets (a) and in target purity (b). Points are the four completed arms at epoch 50 under the frozen mean-pool protocol, one pretraining seed each; the connecting line is a visual guide, not a fit. COVER $f$=.21 is shown as a dashed marker at its measured composition because its epoch-50 probe has not yet run. Policy changes also alter target shape, target count, context budget, and guide source, so this is an observational dose-response hypothesis, not an identified optimum.

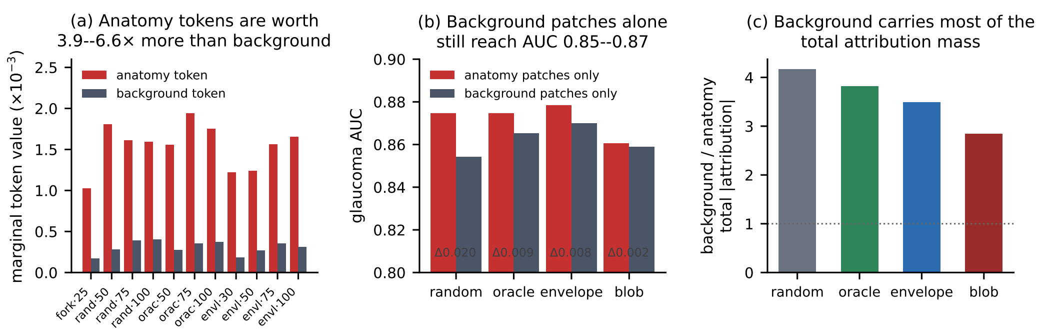


Figure 4: Anatomy is worth more per token, but background is not inert. (a) Marginal context-token value, computed as the increase in prediction error per removed token, for every non-collapsed checkpoint. (b) Region-restricted frozen probes on a separate 2,000/600/1,000 volume split: background positions alone still reach AUC 0.854–0.870, within 0.002–0.020 of anatomy positions. (c) Exact linear attribution: background patches carry 2.8–4.2× the total absolute attribution mass of anatomy patches, because they are far more numerous. Panels (b) and (c) use different protocols from Table 1 and are not comparable to it in absolute level. ViT tokens mix information globally, so “background position” denotes where the token sits, not that black pixels carry signal.

**Target shape is not isolated.** At epoch 30, the stored seed-42 probes give 0.858274 for the anatomy-shaped encoder versus 0.853917 for envelope. A paired stratified bootstrap of those saved predictions ($B = 2{,}000$, seed 42) gives $+0.00436$ with 95% CI $[+0.00097, +0.00776]$ and $p = 0.014$; this holds both encoders fixed and reports test-volume uncertainty only. Envelope uses hard guides while anatomy uses a soft-guide cache, so the arms differ in more than target shape. A separately configured anatomy-v2/blob continuation scores 0.010679 below envelope at epoch 50 (0.865386 versus 0.876064); because this is not a continuation of the epoch-30 anatomy trajectory, it is not a within-run reversal. We therefore report both checkpoints and make no target-shape superiority claim.

Table 2: Within-blob predictor diagnostics. Marginal value is the prediction error increase per removed anatomy token divided by that per removed background-position token.

| Checkpoint | Full-context error | Anatomy/background value |
|---|---|---|
| ep30 | 0.104868 | 3.529× |
| ep40 | 0.224373 | 0.998× |
| ep50 | 0.271158 | 0.830× |
| ep56 | 0.289461 | 0.737× |

## 6 Mechanism: predictor collapse

The obvious explanation for blob's decline is too little visible anatomy, but the accounting does not support it. Blob has the lowest anatomy fraction in context (6.26%), yet among the guided arms it retains the most anatomy cells on average (9.97, versus 9.28 for COVER and 8.63 for envelope) and has the lowest zero-anatomy rate (1.24%). The rectangle arms deliver more anatomy still (18.25 random, 14.93 oracle), so absolute count does not order the arms by AUC either. Neither mean anatomy-cell count nor zero-anatomy rate alone orders the arms by AUC; this does not rule out effects of anatomy fraction, spatial placement, or interactions with target design.

Table 2 gives the stronger within-arm diagnostic. Prediction error rises 2.76× from epoch 30 to 56, while preferential reliance on anatomy context disappears. The intervention also creates a narrower task: blob supplies 64 target slots and 54.744 unique cells, versus 154.624 slots and 118.314 unique cells for envelope, and pads shortfalls with replacement. Near-pure targets, connected geometry, target count, padding, and larger context are not separately identified; jointly they are consistent with, but do not establish, drift toward a positional or prototype-based solution.

**Background positions are not filler.** Figure 4(b,c) quantifies the other side of the trade. Under the region-restricted protocol, envelope epoch-50 background-position pooling reaches AUC 0.870075 against 0.872991 using all positions, and every arm's background-only probe lands within 0.0203 of its anatomy-only probe. In exact linear attribution, background patches carry 2.84–4.16× the total absolute attribution of anatomy patches. Blob is the extreme case in both views: it has the smallest anatomy-over-background probe gap of any arm (0.0016, against 0.0203 for random), and it is the only arm whose background contribution separates glaucoma better than its anatomy contribution (0.855678 versus 0.846425) despite assigning greater per-patch influence to anatomy. The arm that concentrated targets on anatomy most aggressively ended up with the least anatomy-specific representation. Because ViT tokens mix globally, these results locate readable signal at background *positions*; they do not show that black pixels carry glaucoma information, and no eroded-background or content-shuffle control has been run.

## 7 A coverage floor as the balance point

The two preceding sections impose opposing constraints. Concentrating targets on anatomy is what makes the prediction problem informative, but concentrating them exhaustively removes the anchoring context the predictor needs, and the arm collapses. What is required is a policy that hides most anatomy while guaranteeing that some always survives into context.

COVER parameterises exactly this trade with one knob. A coverage floor $f$ acts as a soft stopping rule for adding anatomy-covering blocks (`cover_leave_frac`) and simultaneously as a hard per-slice constraint that at least a fraction $f$ of anatomy cells, and at least four cells, remain visible to the encoder (`cover_min_visible_frac`, `cover_min_visible_cells`). Blocks not needed for coverage are placed as plain uniform rectangles, restricted to windows that preserve the floor.

Figure 5 sweeps $f$ from 0.15 to 0.30 over the 6,137-slice audit. The trade is monotone and well behaved: anatomy placed in targets falls from 78.84% to 64.25% while anatomy reaching context rises from 10.41% to 20.69%, and the blank-context rate falls from 10.74% to 5.51%. Paired against the $f$=.15 reference on the same slices, every higher floor repairs more slices than it breaks; at $f$=.21 the counts are 351 repaired against 173 broken ($z = -7.78$).

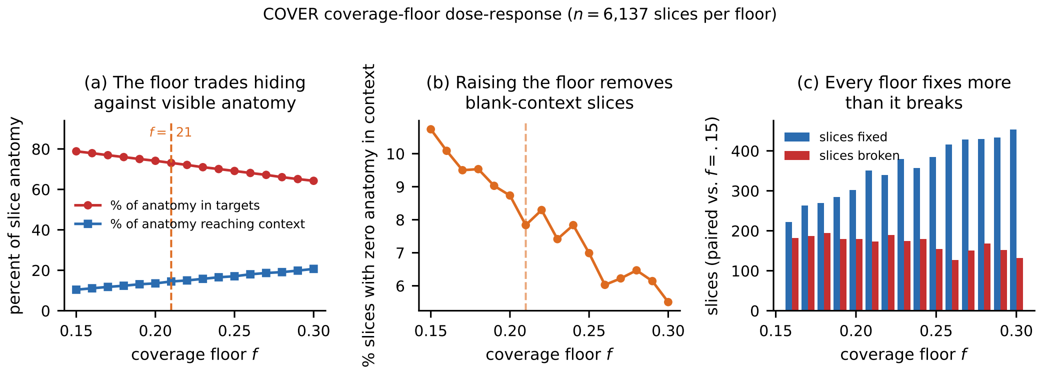


Figure 5: Coverage-floor dose-response, measured by resampling masks over the same 6,137 slices at each floor; no pretraining is involved. (a) Raising $f$ monotonically moves anatomy out of targets and into context. (b) Blank-context slices fall from 10.74% to 5.51%. (c) Against the $f$=.15 reference, every higher floor repairs more slices than it breaks. This sweep identifies the composition trade only; it is not a downstream sweep, and no AUC optimum over $f$ is claimed.

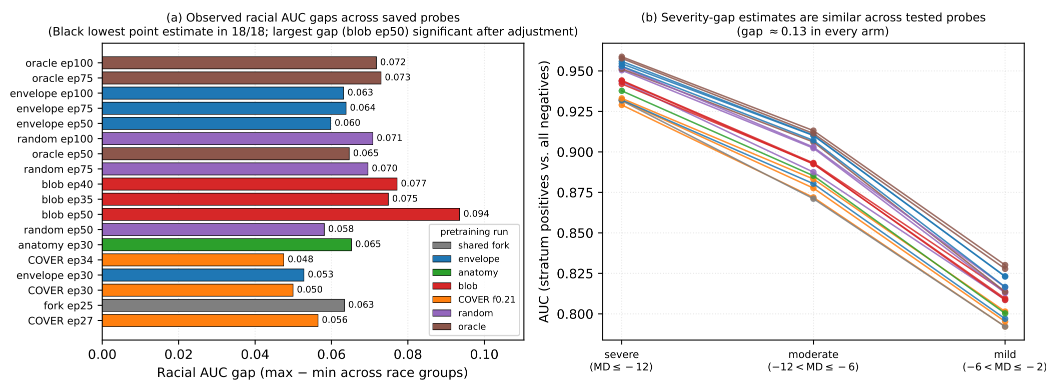


Figure 6: Subgroup results for 18 valid probes from six pretraining trajectories. Left: racial AUC gap by arm; Black patients have the lowest AUC in every probe. Right: each positive severity stratum is compared with the shared negative pool because mean deviation defines the label. One individual racial gap (blob epoch 50, Asian−Black) is significant after multiplicity adjustment; the remainder are not.

Two limits must be stated plainly. First, this is a mask-statistics sweep, not a downstream sweep: it establishes that the floor controls composition in the intended direction, and it cannot locate an AUC-optimal $f$. Second, only $f$=.21 has been pretrained, and its epoch-50 probe had not completed at submission time, so the arm appears in Table 1 and Figure 3 without an AUC. **Assumption, flagged:** the design rationale above predicts that a floored policy should avoid blob's collapse while retaining envelope-like composition, but we do not yet have the number that would test this, and nothing in this paper should be read as reporting it. An earlier archived COVER run reached 0.8558–0.8612 over epochs 26–100, but it was executed with a different target-precision setting than the arms it would be compared against, so its deficit is not attributable and we exclude it.

## 8 Equity and disease severity

Saved test predictions contain no identifiers, so we join them to demographics in deterministic test-loader order. The join is validated by exact label reconstruction, filename order, in-volume race/gender fields, and in-volume labels: all 3,000 labels agree in every probe directory. Four contaminated probes are excluded below, leaving 18 valid probes drawn from six pretraining trajectories.

Table 3: Frozen-readout and regional controls. The regional probes use a separate 2,000/600/1,000 train/validation/test split, so they are not directly interchangeable with the 3,000-volume standard probes.

| Control | Condition | Test AUC |
|---|---|---|
| Probe head, random ep100 | attentive $d = 1$ | 0.8706 |
| | mean-pool | 0.8746 |
| Region, random ep50 | all positions | 0.8608341 |
| | anatomy positions | 0.8746555 |
| | background positions | 0.8543614 |
| Region, oracle ep50 | all positions | 0.8682588 |
| | anatomy positions | 0.8746475 |
| | background positions | 0.8652265 |
| Long-horizon placement | random ep100 | 0.8745809 |
| | oracle ep100 | 0.8854852 |

The test split contains 2,318 white, 431 Black, and 251 Asian participants. Black participants have the lowest AUC in 18/18 valid probes (Figure 6). The max–min racial gap varies $1.97\times$, from 0.0475 for COVER epoch 34 to 0.0935 for blob epoch 50, while overall AUCs differ by 0.0371. The rank correlation between overall AUC and racial gap is Spearman $\rho = 0.488$ (nominal $p = 0.0399$), but the probes are not independent: several are checkpoints of the same pretraining trajectory, so this nominal significance does not license a policy-level inference and we claim no accuracy–fairness tradeoff. One individual gap does reach significance after multiplicity adjustment (blob epoch 50, Asian−Black, 0.0935, $p < 0.05$); the remaining gaps do not. The most robust result is therefore the consistency of the ordering rather than any single gap.

Mean deviation defines the FairVision glaucoma label, making within-bin AUC undefined. We instead compare positives in each severity stratum with the same 1,534 negatives. The severe-to-mild AUC gap is 0.1286–0.1394 in every valid probe, a spread of 0.0108. Descriptively, no tested mask policy narrows the mild-disease penalty; with one trajectory per policy we do not test this causally.

## 9 Ablations and controls

**Batch size and composition.** The $B = 1 \to 64$ comparison in Figure 2 exercises the collation mechanism: zero-anatomy context rises 0.00→4.63% for random, 1.56→10.10% for envelope, and 2.34→11.02% for COVER $f = .15$. The two batch settings are separate audit samples rather than a paired intervention, so they bound the effect without isolating it. Table 1 reports post-collation composition observationally across arms that differ in more than purity, and Figure 3 is the completed four-point purity sweep. A clean COVER $f = .21$ trajectory reaches AUC 0.8483, 0.8522, and 0.8571 at epochs 27, 30, and 34; these are checkpoints within one run, not a cross-floor dose response. The matching epoch-50 probe for this trajectory was still pretraining at the time of writing and is therefore not reported.

**Readout and region controls.** The head ablation shows that mean pooling is not uniquely privileged: a single-query attentive probe scores within 0.004 of it. In the separate random- and oracle-epoch-50 regional probes, anatomy positions outperform background positions; these are arm-specific controls, not a global regional estimate.[1] The epoch-100 oracle/random difference is a descriptive point estimate consistent with anatomical placement helping, but with one trajectory per policy it does not identify a causal policy effect.

## 10 Limitations and confound ledger

These limitations bound every contribution.

[1]The regional ablation uses separately trained probes. Its random/oracle all-position AUCs are 0.8608341/0.8682588 with early stopping at epochs 43/42, whereas the composition table uses standard mean-pool AUCs 0.8640971/0.8740299 stopped at epochs 46/47. Absolute AUCs across the two protocols are not interchangeable.

1. There is one pretraining seed per policy and no pretraining-seed replication anywhere in the programme. The 18 valid subgroup probes arise from only six pretraining trajectories and are non-independent, since several are checkpoints of the same run. The epoch-30 interval is a paired bootstrap over test volumes with both encoders held fixed, not a seed-variance estimate. The random/oracle epoch-50 mean-pool encoders were produced on another machine and reference checkpoints that are unavailable locally, so their arms cannot be locally re-probed or extended to seed replication.
2. The anatomy/envelope epoch-30 comparison differs in continuation history and uses soft versus hard MIRAGE guide caches. At epoch 50, guide, geometry, hidden fraction, target count, padding, and context budget move together; the sign reversal is not a clean shape ablation.
3. Four earlier COVER-random probes are retracted and excluded: their run used `enc_truncate: window` and `amp_target: true`. They cannot be pooled with the clean COVER $f = .21$ trajectory.
4. Blob composition uses $n = 1{,}534$, whereas the other composition rows use $n = 6{,}137$. The purity analysis has only four completed AUC points on one dataset, so it cannot establish an optimum.
5. Predictor collapse is diagnostic, not causal. Purity, target count, connected geometry, replacement padding, and context size change together. Background-only pooling concerns globally mixed background-position tokens, not proof of biomarkers in black pixels.
6. Fairness estimates reuse one test split. Black ($n = 431$) and Asian ($n = 251$) groups are small; only one individual gap reaches significance after multiplicity adjustment (blob epoch 50, Asian−Black), and the nominally significant AUC–gap correlation pseudoreplicates checkpoints drawn from shared trajectories. Policy is confounded with seed and epoch; no accuracy–fairness tradeoff is identified.
7. The study uses one dataset and frozen probes, with historical reuse of its test split and no external cohort. Results may change under end-to-end fine-tuning, other OCT devices, or other anatomy.

**Evidence still required.** Three measurements would materially strengthen these conclusions and are not yet available. First, every policy is represented by exactly one pretraining trajectory; each would have to be repeated across pretraining seeds, with the continuation rather than the probe as the unit of inference, before any policy ranking could be treated as an effect rather than an observation. Second, the anatomy-shaped and envelope continuations should be repeated under a common guide cache, hidden budget, and target-count policy before any shape comparison is attempted. Third, matched epoch-50 AUC across COVER visibility floors is needed before the floor curve can be related to representation quality.

## 11 Conclusion

Masking is not merely a pretraining hyperparameter: batching can silently change its spatial meaning. In this OCT study, stock-style prefix truncation erases lower-row context, target purity relates non-monotonically to downstream quality, a near-pure target policy develops predictor collapse, and subgroup ordering persists despite policy changes. Medical predictive self-supervision should report target composition and the context actually delivered after collation. The present evidence supports that audit standard, not anatomy-shape superiority.

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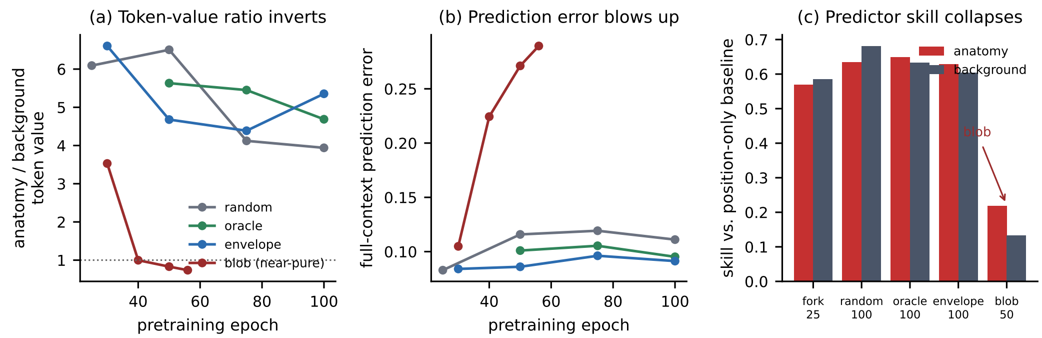


Figure 7: Within-arm collapse diagnostics for the near-pure-anatomy policy. (a) The ratio of anatomy to background marginal token value falls below 1.0 by epoch 40 for blob and stays there, while every other family remains between 3.9 and 6.6. (b) Full-context prediction error rises 2.76× from epoch 30 to 56 for blob and is flat elsewhere. (c) Skill relative to a position-only baseline, measured on 108 held-out slices: blob retains 0.13 on background and 0.22 on anatomy, against 0.57–0.68 for all other checkpoints. Checkpoints are at different epochs across families, so panel (c) compares endpoints rather than a matched budget.

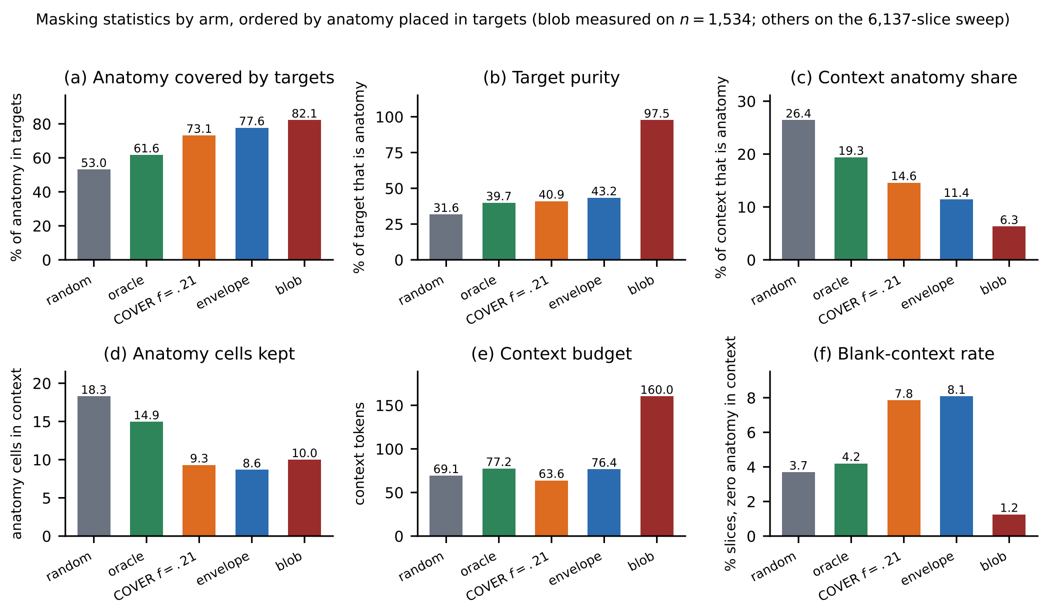


Figure 8: Complete post-collation masking statistics for all five arms, ordered by the share of slice anatomy placed in targets. These are the same values tabulated in Table 1. Note that the two quantities most often assumed to drive representation quality move in opposite directions across the arms: blob has both the lowest context anatomy share (6.3%) and the lowest blank-context rate (1.2%), which is why anatomy starvation does not by itself order the arms by AUC.

# A Supplementary figures

# B Planned extension: dense transfer

**None of the following has been run. No result, number, or direction reported in this section is a measurement; the section states a design and a falsifiable prediction so that the outcome cannot be reinterpreted after the fact.**

The audit in this paper evaluates target composition through frozen linear probes on a single volume-level classification task. That choice isolates representation quality from decoder capacity, but it is also the setting least likely to reward target geometry: a volume-level label can be recovered from pooled features that need not preserve where anatomy is. Our anatomy-shaped arm was not better at classification, and we decline to argue that it should have been.

Dense prediction is the setting where target geometry has a mechanism to matter, because the pretext target and the downstream output share a coordinate system. The planned test is therefore a transfer to layer segmentation:

1. Take the terminal encoder of each pretraining arm already reported here — random, oracle, COVER, envelope, and the anatomy-shaped arm — with no additional pretraining, so the comparison inherits the compute-matched schedule and the warm-start structure documented in the experimental setup.
2. Attach an identical lightweight decoder to every arm and fit it on an external labelled retinal-layer segmentation set that was not used anywhere in this study. Both frozen-encoder and full fine-tuning regimes are run, because the two answer different questions: frozen measures what the pretraining objective already encoded, fine-tuned measures what it made easy to reach.
3. Report Dice and boundary error per layer, across the same race and sex strata used in the equity analysis, with cell sizes printed and bootstrap intervals over volumes.

**Stated prediction, and what would falsify it.** We predict that the ordering observed in this paper reverses under dense transfer: that anatomy-shaped targets, which did not help volume-level classification, will yield higher segmentation Dice than rectangular targets at matched compute, and that the collation repair and visibility floor introduced here will matter more rather than less, because a truncated context removes exactly the rows a segmentation decoder must label. The prediction is falsified if anatomy-shaped and rectangular arms are within overlapping intervals on Dice, which would establish that target geometry is not transferable and that mask shape is a pretraining implementation detail rather than a representational choice. We regard that outcome as fully possible, and it would be reported.

**Why it is not in this paper.** The segmentation labels required are external to the cohort audited here and the transfer runs have not been performed. Reporting the design without the result is deliberate: the composition audit, the collation defect, and the subgroup ordering stand on the evidence already presented, and none of them depends on how the dense transfer turns out.